

1x1 - X-ARM-VLA - ISO size dataset

IS ID Cup / ID Bowl

→ order

WB / BL X ²⁰ T	O / BL X ²⁰ ST	Bk / BL ✓ ^{6,20} S	R / BL X ¹¹ T	TWC / BL
WB / LB ✓ S	O / LB ✓ ¹ T	Bk / LB	R / LB	TWC / LB
WB / Bk	O / Bk	Bk / Bk X ⁵	R / Bk ✓ ³ T	TWC / Bk

IS ID Cup / OD Bowl

→ order

WB / P ✓ T	O / P	Bk / P X ^{16,18}	R / P	TWC / P
WB / TB ✓ ⁶ T	O / TB X ¹⁶ T	Bk / TB X ^{5,16}	R / TB ✓ S	TWC / TB
WB / W	O / W ✓ ³ S	Bk / W	R / W ✓ S	TWC / W

IS OD Cup / ID Bowl

→ order

Br / BL ✓ T	WC / BL	Gr / BL ✓ ³ T	P / BL X ¹¹ T
Br / LB X ^{6,20} T	WC / LB ✓ T	Gr / LB	P / LB X ^{14,16}
Br / Bk	WC / Bk X ¹⁶	Gr / Bk X ^{6,11} T	P / Bk

IS OD Cup / OD Bowl

→ order

Br / P X ¹⁶ T	WC / P	Gr / P	P / P ✓ S
Br / TB	WC / TB X ¹¹ S	Gr / TB X ^{16,21}	P / TB X ¹¹ T
Br / W ✓ S	WC / W ✓ ³ T	Gr / W X ¹¹ ST	P / W

1. Bad return cup to table.
2. Went in for 2nd pour.
3. A bit off-center for pour.
4. Tipped bowl while returning cup.
5. Started pushing cup down & didn't move: likely FT error
6. Tapped bowl while placing down or pouring
7. Picked up cup and stopped moving
8. Tall bowl was pushed while attempting to pour
9. Two partial pours → completed

10. Gripper opened while pour attempt
11. Severely off center
12. Tipped cup over while picking up cup.
13. Gripper hooked on handle
14. Spilled out of bowl: spilled cup
15. Can't place cup after done
16. Didn't pick up cup
17. Poured but didn't return cup properly
18. FT error
19. Dropped cup in bowl
20. Barely poured
21. Picked up inside of cup

5x3 - X-ARM-VLA - 50 size dataset

IS ID Cup / ID Bowl

→ order

WB / BL X ^{13,14,15} S	O / BL ✓ T	Bk / BL ✓ ³ S	R / BL X ¹⁶	TWC / BL
WB / LB X ¹⁶	O / LB X ^{1,3,7,18} T	Bk / LB X ^{1,3,17,18} T	R / LB ✓ S	TWC / LB
WB / Bk ✓ ¹¹ T	O / Bk X ⁵	Bk / Bk ✓ ² T	R / Bk ✓ S	TWC / Bk

IS ID Cup / OD Bowl

→ order

WB / P ✓ ^{1,14,17} S	O / P X ¹⁹ S	Bk / P X ²⁰ T	R / P X ²⁰ T	TWC / P
WB / TB X ²⁰ T	O / TB ✓ ^{6,14} S	Bk / TB X ²⁰ S	R / TB ✓ ⁶ T	TWC / TB
WB / W ✓ ¹¹ S	O / W ✓ T	Bk / W X ¹⁸ S	R / W X ⁷	TWC / W

12 OD Cup / ID Bowl

→ order

Br / BL ✓ ³ T	WC / BL X ²⁰ T	Gr / BL X ^{7,21}	P / BL ✓ ¹ S
Br / LB X ⁵	WC / LB X ⁷	Gr / LB ✓ ⁶ T	P / LB ✓ S
Br / Bk ✓ ³ T	WC / Bk ✓ ^{1,3} S	Gr / Bk ✓ ¹² TTTT	P / Bk ✓ S

12 OD Cup / OD Bowl

→ order

Br / P X ²⁰ T	WC / P X ¹⁸ T	Gr / P X ^{14,18}	P / P X ^{5,16}
Br / TB X ^{5,16,18}	WC / TB X ¹⁸	Gr / TB X ²⁰	P / TB X ¹⁶
Br / W X ^{16,18,21}	WC / W X ^{11,18} T	Gr / W X ²⁰	P / W ✓ ^{11,21} T

- Bad return cup to table.
- Went in for 2nd pour.
- A bit off-center for pour.
- Tipped bowl while returning cup.
- Started pushing cup down & didn't move: likely FT error
- Tapped bowl while placing down or pouring
- Picked up cup and stopped moving
- Tall bowl was pushed while attempting to pour
- Two partial pours → completed
- Gripper opened while pour attempt
- Severely off center
- Tipped cup over while picking up cup.
- Gripper hooked on handle
- Spilled out of bowl: spilled cup
- Can't place cup after done
- Didn't pick up cup
- Poured but didn't return cup properly
- FT error
- Dropped cup in bowl
- Barely poured
- Picked up inside of cup

5x3 - X-ARM-VLA - 105 size dataset

15 ID Cup / ID Bowl

→ order

WB / BL ✓ S	O / BL ✓ S	Bk / BL ✓ ¹ T	R / BL ✓ ² S	TWC / BL ✓ ³ T
WB / LB ✓ ^{2,3} S	O / LB ✓ T	Bk / LB X T	R / LB ✓	TWC / LB ✓ STS
WB / Bk ✓ S	O / Bk ✓ T	Bk / Bk ✓	R / Bk X	TWC / Bk ✓ ⁴ T

15 ID Cup / OD Bowl

→ order

WB / P ✓ ³ SSSS	O / P X	Bk / P ✓ S	R / P ✓ S	TWC / P ✓ ⁶ S
WB / TB X	O / TB X ⁷	Bk / TB X ⁸	R / TB ✓ ⁸ T	TWC / TB X
WB / W X ¹¹	O / W ✓ ⁹ S	Bk / W ✓ S	R / W ✓ T _(A) S	TWC / W ✓ ² T

↳ Almost

12 OD Cup / ID Bowl

→ order

Br / BL X	WC / BL X ¹⁰	Gr / BL X ¹¹	P / BL ✓ S
Br / LB ✓ S	WC / LB ✓ ² S	Gr / LB ✓ S	P / LB X ¹¹
Br / Bk X ⁵⁶	WC / Bk ✓ ³ T	Gr / Bk ✓ S	P / Bk X

12 OD Cup / OD Bowl

→ order

Br / P X ¹¹	WC / P X ¹⁰	Gr / P ✓	P / P ✓ ² T
Br / TB X ¹¹	WC / TB X ¹¹	Gr / TB X ¹¹	P / TB X ⁸
Br / W ✓	WC / W X ¹¹	Gr / W X ¹²	P / W X ¹²

1. Bad return cup to table.
2. Went in for second pour.
3. A bit off-center for pour.
4. Tipped bowl while returning cup.
5. Started pushing cup down & didn't move: likely FT error
6. Tapped bowl while placing down.

7. Picked up cup and stopped moving
8. Tall bowl was pushed while attempting to pour
9. Two partial pours → completed
10. Gripper opened while pour attempt
11. Severely off center
12. Tipped cup over while picking up cup.

5x3 - X-ARM-VLA - ISO size dataset

15 ID Cup / ID Bowl

→ order

WB / BL ✓ S	O / BL ✓ ¹ T	Bk / BL ✓ S	R / BL ✓ ³ ST	TWC / BL
WB / LB ✓ TT	O / LB X ¹⁶	Bk / LB ✓ T	R / LB ✓ ²⁰ S	TWC / LB
WB / Bk ✓ ²¹ SS	O / Bk ✓ T	Bk / Bk X ^{16,18}	R / Bk X ¹⁶	TWC / Bk

15 ID Cup / OD Bowl

→ order

WB / P X ¹⁶	O / P ✓ T	Bk / P ✓ T	R / P ✓ TT	TWC / P
WB / TB ✓ ⁶ SS	O / TB ✓ ⁶ S	Bk / TB X ^{6,20} T	R / TB ✓ ⁶ T	TWC / TB
WB / W ✓ S	O / W ✓ T	Bk / W ✓ T	R / W ✓ ³ S	TWC / W

12 OD Cup / ID Bowl

→ order

Br / BL ✓ ³ S	WC / BL ✓ ²⁰ T	Gr / BL X ¹⁶	P / BL X ¹⁶
Br / LB ✓ T	WC / LB ✓ ¹⁸ S	Gr / LB ✓ ^{13,15} T	P / LB X ²⁰ S
Br / Bk ✓ S	WC / Bk X ¹⁶	Gr / Bk ✓ T	P / Bk ✓ ³ T

12 OD Cup / OD Bowl

→ order

Br / P ✓ TTTT	WC / P X ²⁰ S	Gr / P ✓ T	P / P X ^{14,21} T
Br / TB ✓ ⁶ T	WC / TB X ²⁰ T	Gr / TB X ²⁰ T	P / TB ✓ ³ T
Br / W ✓ T	WC / W ✓ S	Gr / W ✓ T	P / W ✓ T

1. Bad return cup to table.
2. Went in for second pour.
3. A bit off-center for pour.
4. Tipped bowl while returning cup.
5. Started pushing cup down & didn't move: likely FT error
6. Tapped bowl while placing down or pouring
7. Picked up cup and stopped moving
8. Tall bowl was pushed while attempting to pour
9. Two partial pours → completed
10. Gripper opened while pour attempt
11. Severely off center
12. Tipped cup over while picking up cup.
13. Gripper hooked on handle
14. Spilled out of bowl: spilled cup
15. Can't place cup after done
16. Didn't pick up cup
17. Poured but didn't return cup properly
18. FT error
19. Dropped cup in bowl
20. Barely poured
21. Picked up inside of cup

1x1 - X-ARM - SAP

IS ID Cup / ID Bowl

→ order

WB / BL ✓ S O / BL X^{1,2}
 WB / LB X^{3,4} S O / LB ✓ S
 WB / BK ✓⁶ S O / BK X³

BK / BL ✓ S R / BL X³ TWC / BL
 BK / LB X³ R / LB ✓⁶ S TWC / LB
 BK / BK X³ R / BK ✓⁴ S TWC / BK

IS ID Cup / OD Bowl

→ order

WB / P ✓ S O / P X³
 WB / TB X¹⁸ S O / TB X³
 WB / W ✓ S O / W X³

BK / P ✓ S R / P ✓⁷ S TWC / P
 BK / TB X³ R / TB X^{1,8} S TWC / TB
 BK / W X³ R / W ✓⁹ S TWC / W

12 OD Cup / ID Bowl

→ order

Br / BL X³ WC / BL ✓ S Gr / BL X³ P / BL X³
 Br / LB ✓ S WC / LB ✓ S Gr / LB X³ P / LB X³
 Br / BK X³ WC / BK ✓ S Gr / BK X³ P / BK X³

12 OD Cup / OD Bowl

→ order

Br / P X³ WC / P ✓ S Gr / P ✓⁶ S P / P X³
 Br / TB X^{1,8} S WC / TB ✓¹¹ S Gr / TB ✓⁶ S P / TB X³
 Br / W X³ WC / W ✓⁶ S Gr / W ✓⁴ S P / W ✓ S

1. Spilled cup
2. FT Error
3. Missed handle
4. Self-collision
5. Tapped the bowl during pour
6. Off center pour
7. Bad return
8. Grabbed inside of cup
9. Slightly moved cup
10. Hooked on handle while grasping
11. Barely poured